

Assignment - 05

Subject: Adv. Java

Date: 01/12/24

• Section - A (5x2=10)

1. Explain life cycle of JSP. Also write 3 life-cycle methods for a JSP.

Ans The life cycle of JSP involves these steps:

(i) Translation and Compilation:- The JSP file is translated into a servlet and then compiled.

(ii) Initialization:- The `jspInit()` method is called once when the JSP is loaded.(iii) Execution:- The `service()` method is called for each request.(iv) Destruction:- The `jspDestroy()` method is called before the JSP is unloaded.

3 Life-Cycle Methods of JSP:-

(i) `jspInit()` - initialise the JSP.(ii) `JSPService(HTTPServletRequest req, HttpServletResponse res)` - Handle request and response.(iii) `jspDestroy()` - Cleans up resource before JSP is unloaded.

2. What is difference b/w include directive and include action?

Ans i) Include Directive (`<%@ include file = "---" %>`):

• Include content at compile time.

• Static inclusion; the included file content becomes part of the JSP.

(ii) Include Action (`<jsp:include page = "---" />`):

• Include content at request time.

• Dynamic inclusion; the included file is processed separately and then included.

3. Give the syntax for JSP comments.

Ans Syntax for JSP comment is:-

`<%-- This is a JSP comment -- %>`

Key points:-

JSP comments are not sent to the client (not visible in the HTML source).

4. List all the directives in JSP.

Ans The three directives in JSP are:-

i) `<%@ Page %>`ii) `<%@ include %>`iii) `<%@ taglib %>`

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5. What are the three tags used in JSP bean development?

Ans The three tags used in JSP bean development are:

- i) `<jsp:useBean>` - To create or locate a bean.
- ii) `<jsp:setProperty>` - To set properties of the bean.
- iii) `<jsp:getProperty>` - To retrieve properties of the bean.

Section-B (3x3=21)

1. List out some advantages of using JSP.

Ans Advantages of using JSP:

- i) Simplified Development :- Allows embedding Java code directly into HTML, making it easy to create dynamic web pages.
- ii) Separation of Concerns :- Supports MVC architecture, enabling separation of presentation logic from business logic.
- iii) Reusable Components - JSP allows use of JavaBean and custom tags, promoting code reusability.
- iv) Platform Independence - Runs on any platform with a compatible servlet container (e.g. - tomcat).
- v) Built-in Objects: Provides pre defined objects like "request", "response", "session", etc. to simplify web development.
- vi) Efficient Compilation :- JSP are compiled into servlet for faster execution.
- vii) Integration with Java :- Leverages the full power of Java for complex logic and backend integration.

2. List all JSP implicit objects and also write their class name.

Ans JSP Implicit Objects and Their Class Names:-

Implicit Object	Class Name
(i) Out	- <code>JSPWriter</code>
(ii) request	- <code>HttpServletRequest</code>
(iii) response	- <code>HttpServletResponse</code>
(iv) session	- <code>HttpSession</code>
v) config	- <code>Servlet</code>

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- vi) application - ServletContext
- vii) pageContext - PageContext
- viii) page - Object
- ix) exception - Throwable

2. How can we stop errors on Display in a JSP Page? (Only a Program of exception handling using element of Page directive.)

Ans. To stop errors from displaying on a JSP Page, we can use exception handling with the "errorPage" and 'isErrorPage' attributes of the JSP page directive. Here's a demonstration:-

Program : Exception Handling in JSP

- Main Page (main.jsp) -

```
<%@ Page errorPage = "error.jsp" %>
<%!
    public int divide (int a, int b) {
        return a/b;
    }
%>
<html>
<body>
<h2> Exception Handling in JSP </h2>
<%
    int num1 = 10;
    int num2 = 0;
    int result = divide (num1, num2);
%>
<p> Result : <%= result %> </p>
</body>
</html>
```

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Error page (error.jsp)

```

<%@ page isErrorPage = "true" %>
<html>
  <body>
    <h2> Error Handling </h2>
    <p> An error occurred: </p>
    <p><b><%= exception.getMessage() %></b></p>
  </body>
</html>

```

• Explanation -

- i) The `isErrorPage = "true"` in `main.jsp` specifies the page to display if an exception occurs.
- ii) If an error occurs, the control is transferred to `error.jsp` and the error details are displayed.

Output:for `main.jsp`:Exception occurs (e.g., Division by zero), control redirects to `error.jsp`.for `error.jsp`:

Error Handling

An error occurred:

/ by zero.

• Section - C (3x11=33)

1. Write a JSP Program to demonstrate the use of JSP page and include directive.

Ans: JSP Program to demonstrate page and include directive.

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- Main Page (main.jsp)

```
<%@page contentType="text/html" language="java"%>
<%@include file="header.jsp"%> <!-- Static Inclusion
      using Include Directive -->
```

```
<html>
```

```
<body>
```

```
<h2> Welcome to JSP Directives Demo </h2>
```

```
<p> This content is part of the main JSP page. </p>
```

```
<hr>
```

```
<p> Dynamic Content: </p>
```

```
<jsp:include page="footer.jsp" /> <!-- Dynamic
      inclusion using Include action -->
```

```
</body>
```

```
</html>
```

- Header Page (header.jsp)

```
<%@page contentType="text/html" language="java"%>
```

```
<h1> JSP Page and Include Directive Example </h1>
```

```
<hr>
```

- Footer Page (footer.jsp)

```
<%@page contentType="text/html" language="java"%>
```

```
<p> Current Time: <%= new java.util.Date() %> </p>
```

```
<hr>
```

```
<p> © Copy; 2024 JSP Demo </p>
```

Output: JSP Page and Include Directive Example

Welcome to JSP Directives Demo

This content is part of the main JSP page.

Dynamic Content:

Current Time: Sun Dec 01 15:45:00 IST 2024

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2. Write a JSP program to demonstrate the declaration, scriptlet and expressions.

Ans. JSP Program to demonstrate Declaration, Scriptlet, and Expressions.

Program: declarationSE.jsp

```
<%@ page contentType="text/html" language="java" %>
<html>
  <head>
    <title> JSP Declaration, Scriptlet, and Expression </title>
  </head>
  <body>
    <h1> JSP Declaration, Scriptlet, and Expression Example </h1>

    <!-- Declaration -->
    <%
      int addNumbers(int a, int b) {
        return a+b;
      }
    %>

    <!-- Scriptlet -->
    <%
      int num1 = 10;
      int num2 = 20;
      int sum = addNumbers(num1, num2);
    %>

    <p> Scriptlet : The sum of <%= num1 %> and <%= num2 %>
    is calculated in the scriptlet. </p>

    <!-- Expression -->
    <p> Expression: The calculated sum is: <%= sum %> </p>
  </body>
</html>
```

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Output:

JSP Declaration, Scriptlet, and Expression Example

Scriptlet: The sum of 10 and 20 is calculated in the scriptlet.

Expression: The calculated sum is : 30

Q3. Write a complete JSP program to create a custom tag `<my:jerc></my:jerc>`. This tag will display the name and address of the JERC University.

Ans. Steps to create and use a custom JSP tag.

1. Tag Handler Class (JercTagHandler.java)

Create a Java class to define the behaviour of the `<my:jerc>` tag.

```
package mytags;
import javax.servlet.jsp.JspException;
import javax.servlet.jsp.JspWriter;
import javax.servlet.jsp.tagext.SimpleTagSupport;
```

```
public class JercTagHandler extends SimpleTagSupport {
```

```
    @Override
```

```
    public void doTag() throws JspException, IOException {
```

```
        JspWriter out = getJspContext().getOut();
```

```
        out.println("<h3> JERC University </h3>");
```

```
        out.println("<p> Address : Plot No. IS-2036 to 2039,
```

```
        Ramchandrapur, Sitapur Industrial area, Jaipur, Rajasthan
```

```
        - 303905 </p>");
```

```
    }
```

```
}
```

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2. Tag Library Descriptor (TLD)

Create a 'tags.tld' file to define the custom tag.

web.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<taglib xmlns="http://java.sun.com/xml/ns/javac"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://java.sun.com/xml/ns/javac
        http://java.sun.com/xml/ns/javac/web-isptaglibrary-2-1.xsd"
        version="2.1">
```

```
<tlb-version>1.0</tlb-version>
```

```
<short-name>Mytags</short-name>
```

```
<uri>http://mytags</uri>
```

```
<tag>
```

```
<name>jecre</name>
```

```
<tag-class>mytags.JecreTagHandler</tag-class>
```

```
<body-content>empty</body-content>
```

```
</tag>
```

```
</taglib>
```

3. JSP File: - (Custom Tag Example.jsp)

Create a JSP file to use the custom tag.

```
<%@taglib prefix="my" uri="http://mytags"%>
```

```
<html>
```

```
<head>
```

```
<title>Custom JSP Tag Example</title>
```

```
</head>
```

```
<body>
```

```
<h1>Using custom JSP Tags</h1>
```

22BCON1068

Ankit Ku

Don't forget to
activate Office 2021.
To begin, open
an Office app.

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```
<!-- Custom Tag -->  
<my:jecrc />  
</body>  
</html>
```

O/P:

Using Custom JSP Tags

JECRC University

Address: Plot No. IS-2036 to 2039, Ramchandrapura, Sitapura
Industrial Area, Jaipur, Rajasthan - 303905

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